

Ranking Visible Content by CTR Review Priority: A Decision-Support Model for Engagement Opportunity Scoring

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Abstract

Search teams have limited time to manually review content for click-through-rate (CTR) issues, so a ranked queue that surfaces the most promising review candidates first has direct operational value. This paper asks whether content likely to under-capture clicks in a future month can be identified using only information available before that month. Using pseudonymized client/content-level Google Search Console aggregates from the FlyRank internship warehouse, five pre-outcome features from a February 2026 feature window (log impressions, log clicks, CTR, average position, and days with impressions) are used to predict whether a page’s March 2026 CTR falls below the median CTR of an eligible evaluation population. A logistic regression model is compared against a transparent Week-4 rule-based baseline on an identical, client-grouped held-out test split of 13,017 rows. The model reaches an Average Precision of 0.912 and ROC AUC of 0.888, against a baseline of 0.828 AP and 0.781 AUC – an improvement of +0.084 AP and +0.107 AUC on the same split. The result is observational and decision-support only: it ranks a CTR review queue, it does not establish a causal effect on click-through behavior.

Lane: CTR / Engagement Opportunity Scoring

1 Introduction / Problem Statement

Content and SEO teams routinely face more pages that could plausibly benefit from a CTR review (title, snippet, or metadata changes) than they have time to review manually. Without a prioritization signal, review effort is often spent in whatever order pages are noticed, which is not necessarily the order of expected value.

This work supports a single decision: *given a set of pages that already have meaningful search visibility, which ones should a content/SEO team look at first for CTR review?* The unit of analysis is one client–content pair. The output is a ranked queue, each row carrying a priority score, a reason code, and a recommended action (CTR_REVIEW or MONITOR). The cost of a wrong call is bounded and asymmetric: a false positive costs a reviewer a few minutes looking at a page that turns out to be fine, while a false negative means a genuinely under-performing page waits longer to be noticed. This makes a ranking-first approach – optimizing for how well the queue orders true opportunities near the top – more appropriate than a hard classification threshold, and motivates Average Precision as the primary evaluation metric.

Data and machine learning help here because the pattern to be captured (which combinations of prior volume, prior CTR, and prior position tend to precede continued low CTR) is a soft, multivariate signal that is easy for a simple weighted rule to approximate crudely, but that a model can combine more effectively – while still remaining fully transparent about its five inputs.

2 Data

The analysis uses the `fact_content_daily_performance` table from the FlyRank internship warehouse, aggregated to the `client_hash_id × content_hash_id` grain.

- **Feature window:** February 2026.
- **Outcome window:** March 2026 (used only to define the evaluation label, never as a model feature).
- **Fields used:** GSC impressions, GSC clicks, GSC (impression-weighted) average position, and GSC data-availability flag.
- **Excluded:** the `_sample` partition, all future/March-derived fields from the feature set, query-level outcome data, and any client, domain, or query identifiers beyond the pseudonymous hashed IDs used strictly for grouping.

After filtering to rows with available GSC data and at least 100 February impressions and 3 February clicks, the February feature table contains **29,729** client–content rows. The March evaluation population – rows with at least 100 March impressions and 3 March clicks – contains **38,459** rows. Joining the two on the client/content key and dropping rows with missing feature values yields the final modeling table of **24,896** rows, with a positive rate (low-CTR-in-March) of **53.8%**.

No client names, domains, URLs, private queries, or credentials appear anywhere in this paper, the underlying notebook, or its outputs.

3 Methodology

3.1 Features

Five features, all computable from data available strictly before the March outcome window, are used by both the baseline and the model:

1. `log_imp_feb` – $\log(1 + \text{February impressions})$
2. `log_clk_feb` – $\log(1 + \text{February clicks})$
3. `ctr_feb` – February clicks / February impressions
4. `pos_feb` – February impression-weighted average search position
5. `days_withimps_feb` – number of February days with non-zero impressions

3.2 Label

Among content in the eligible March evaluation population (≥ 100 impressions, ≥ 3 clicks), `low_ctr_mar` = 1 when a row’s March CTR falls below the median March CTR of that same eligible population (median cutoff = 0.0035), and 0 otherwise.

3.3 Baseline: the Week-4 rule

A transparent, non-learned scoring rule combines three ranked (percentile) components with fixed weights:

$$\text{baseline_score} = 0.45 \cdot \text{volume_pct} + 0.35 \cdot (1 - \text{ctr_pct}) + 0.20 \cdot (1 - \text{position_pct})$$

i.e. higher February volume, lower February CTR, and better (numerically lower) February position each push a page’s review priority up. This rule uses exactly the same underlying signals as the model and requires no training, making it a fair and interpretable point of comparison.

3.4 Model

A logistic regression classifier (`scikit-learn`), wrapped in a pipeline of median imputation and standard scaling, is trained on the same five features to predict `low_ctr_mar`. Its predicted probability is used directly as the ranking score.

3.5 Validation

Data is split by `client_hash_id` using a grouped shuffle split (75/25, random state 42), so no client’s content appears in both the training and test sets. This guards against a client-level leakage path where within-client similarity would otherwise inflate held-out performance. The baseline and the model are scored on the identical test rows: **11,879** training rows and **13,017** test rows.

3.6 Leakage check

None of `imp_mar`, `clk_mar`, `ctr_mar`, `low_ctr_mar`, or any other March-derived field is present in `feature_cols`. This is verified programmatically in the notebook with an explicit disjointness assertion between `feature_cols` and the set of outcome-only columns, and the check passes.

4 Results

Table 1 reports both methods on the identical grouped test split. Average Precision (AP) is the primary metric, since the deliverable is a ranked queue and the positive class (the lower-CTR opportunity) is the class of interest; ROC AUC is reported as a secondary ranking metric. The base rate (majority-class share) on this split is 53.8% positive, so both metrics are reported against that context rather than in isolation.

Table 1: Model vs. baseline on the same held-out, client-grouped test split (n = 13,017).

Method	Average Precision	ROC AUC	Test rows
Week-4 rule baseline	0.828	0.781	13,017
Logistic regression	0.912	0.888	13,017
Δ (model – baseline)	+0.084	+0.107	–

The logistic regression model ranks held-out low-CTR opportunities more effectively than the Week-4 rule on this split, on both metrics. Figure 2 shows visible separation between the score distributions of the two March-label groups, with the positive-label group shifted toward higher predicted probabilities, consistent with the AP/AUC improvement. This is an association/prediction result on a single held-out split; it does not establish causality or explain how any search engine’s ranking system behaves.

5 Limitations

- The result is **observational and directional**. It does not demonstrate that changing a page’s title, snippet, or content will cause its CTR to change.
- The label is defined relative to the March outcome window’s own median CTR, so the model reflects this evaluation setup rather than a universal, time-invariant definition of an “under-performing” page.
- GSC availability can vary row to row; rows without available GSC data are excluded from all aggregates, which may not be a uniformly random exclusion.

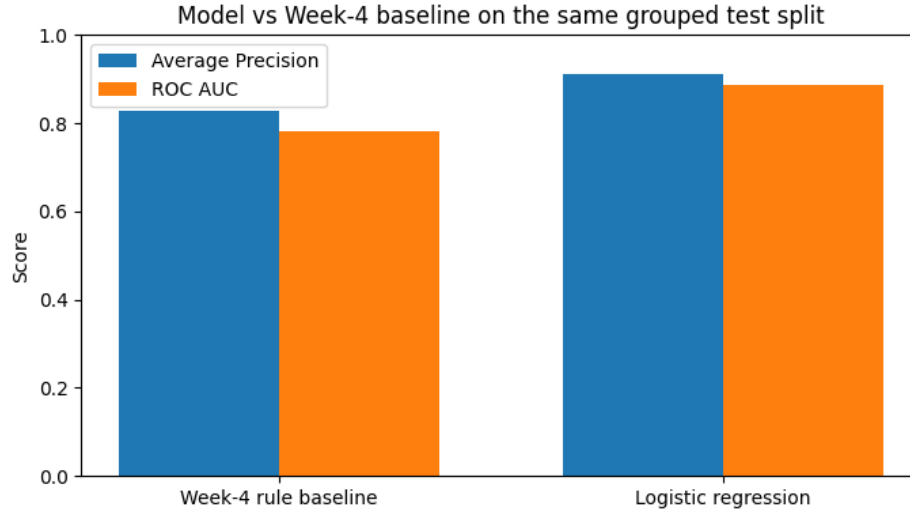


Figure 1: Average Precision and ROC AUC for the Week-4 baseline versus the logistic regression model, evaluated on the same grouped test split.

- The model intentionally uses a compact, five-feature, fully transparent input set. It is designed as an auditable baseline model, not a production-grade ranking system.
- The client-grouped validation split controls for client overlap between train and test, but it is not a true prospective (future-period) holdout, and results may not generalize to months structurally different from February–March 2026.
- All reported artifacts are client-safe by construction: no client names, domains, URLs, private queries, credentials, or raw exports appear in the notebook, its outputs, or this paper.

6 Ranked Recommendations

The trained model’s predicted probability is used as the final priority score for every row in the held-out test set, producing a ranked review queue. Each row carries:

- **Score** – the model’s predicted probability of being a low-CTR opportunity.
- **Reason code** – `VISIBLELOW_CTR_OPPORTUNITY` (intentionally broad and deterministic: the queue is scoped to already-visible content where CTR is the review question).
- **Action** – `CTR_REVIEW` for the top quartile of scores (recommended for prioritized human review), `MONITOR` otherwise (kept in the monitoring queue rather than prioritized).

In practice, a FlyRank content/SEO editor would start each review cycle at the top of this queue, working down through the `CTR_REVIEW`-flagged rows first, using the existing February volume, CTR, and position figures attached to each row as immediate context for the review, and treating `MONITOR` rows as lower priority rather than as pages with no issue. The full queue is exported to `work/outputs/` as `capstone_ranked_recommendations.csv`.

7 Reproducibility

- **Notebook:** `work/notebooks/capstone.ipynb` in the linked repository reproduces every number, table, and figure in this paper, end to end.

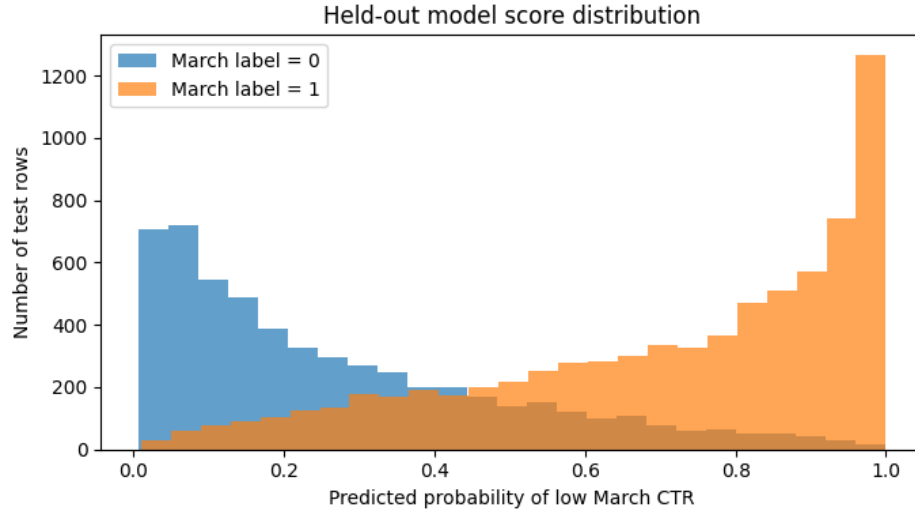


Figure 2: Held-out distribution of the model’s predicted probability of low March CTR, split by the true March label.

- **Environment:** Python 3, duckdb, pandas, numpy, scikit-learn, matplotlib. Data is read directly from the gated Hugging Face warehouse via DuckDB’s `hf://` interface; an `HF_TOKEN` with read access is required and is never written into the notebook.
- **Random seed:** `random_state = 42` for both the grouped train/test split and the logistic regression fit.
- **Split:** `GroupShuffleSplit` with a single split, 25% test size, and `random_state=42`, grouped by `client_hash_id`.
- **Repository:** <https://github.com/abdelrahmanmohamed05/FlyRank-AI/>

Acknowledgments & Data Credit

Built on the FlyRank ML Internship dataset. Data courtesy of [FlyRank](#).